Software and Hardware Tools

Q1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Ans:

Procedure :

1. Download and install Wireshark.
2. Open Wireshark and select the active Wi-Fi network interface.
3. Start packet capture.
4. Browse YouTube for approximately 2 minutes.
5. Stop the packet capture.
6. Save the captured traffic as Youtube_Capture.pcapng.
7. Open Statistics → Protocol Hierarchy.
8. Identify the top three protocols observed during the capture.

Figure 1: Wireshark Capturing Network Traffic While Browsing YouTube

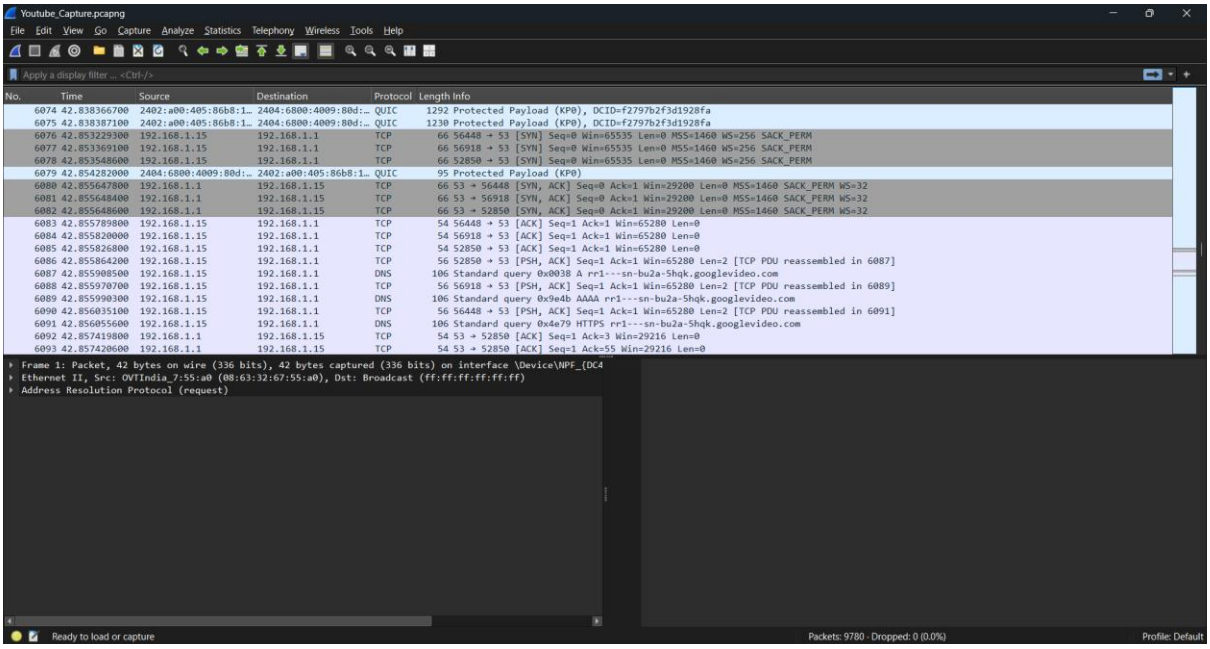


Figure 2: Protocol Hierarchy Showing Captured Network Protocols

Wireshark · Protocol Hierarchy Statistics · Youtube_Capture.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	9780	100.0	8209205	1072 k	0	0	0	9780
Ethernet	100.0	9780	1.7	136920	17 k	0	0	0	9780
Internet Protocol Version 6	87.5	8557	4.2	342280	44 k	0	0	0	8557
User Datagram Protocol	83.4	8160	0.8	65280	8528	0	0	0	8160
QUIC IETF	82.3	8047	88.7	7282388	951 k	8047	7269588	949 k	8077
Multicast Domain Name System	0.0	3	0.0	183	23	3	183	23	3
Domain Name System	0.0	4	0.0	644	84	4	644	84	4
Data	1.1	106	0.6	51311	6703	106	51311	6703	106
Transmission Control Protocol	3.9	384	0.1	8112	1059	190	4232	552	384
Transport Layer Security	1.9	182	2.5	205381	26 k	182	204161	26 k	183
Domain Name System	0.0	4	0.0	276	36	4	276	36	4
Data	0.1	8	0.0	8	1	8	8	1	8
Internet Control Message Protocol v6	0.1	13	0.0	416	54	13	416	54	13
Internet Protocol Version 4	11.9	1160	0.3	23200	3030	0	0	0	1160
User Datagram Protocol	0.4	44	0.0	352	45	0	0	0	44
Simple Service Discovery Protocol	0.0	1	0.0	125	16	1	125	16	1
QUIC IETF	0.4	36	0.2	19153	2502	36	18973	2478	38
Multicast Domain Name System	0.0	3	0.0	183	23	3	183	23	3
Data	0.0	4	0.0	354	46	4	354	46	4
Transmission Control Protocol	11.4	1114	0.3	24356	3181	931	20696	2703	1114
Transport Layer Security	0.2	24	0.4	32828	4288	24	32828	4288	24
Domain Name System	1.6	158	0.1	12282	1604	158	12282	1604	158
Data	0.0	1	0.0	1	0	1	1	0	1
Internet Control Message Protocol	0.0	2	0.0	128	16	2	128	16	2
Address Resolution Protocol	0.6	63	0.0	1764	230	63	1764	230	63
802.1Q Virtual LAN	9.1	892	0.0	3568	466	0	0	0	892

No display filter.

Close | Copy | Protocols | Help

Top Three Protocols Observed

Protocol	**Purpose**
QUIC	Provides fast and secure communication for web applications such as YouTube.
TCP	Ensures reliable transmission of data between devices.
DNS (Domain Name System)	Converts website names into IP addresses before communication begins.

Explanation

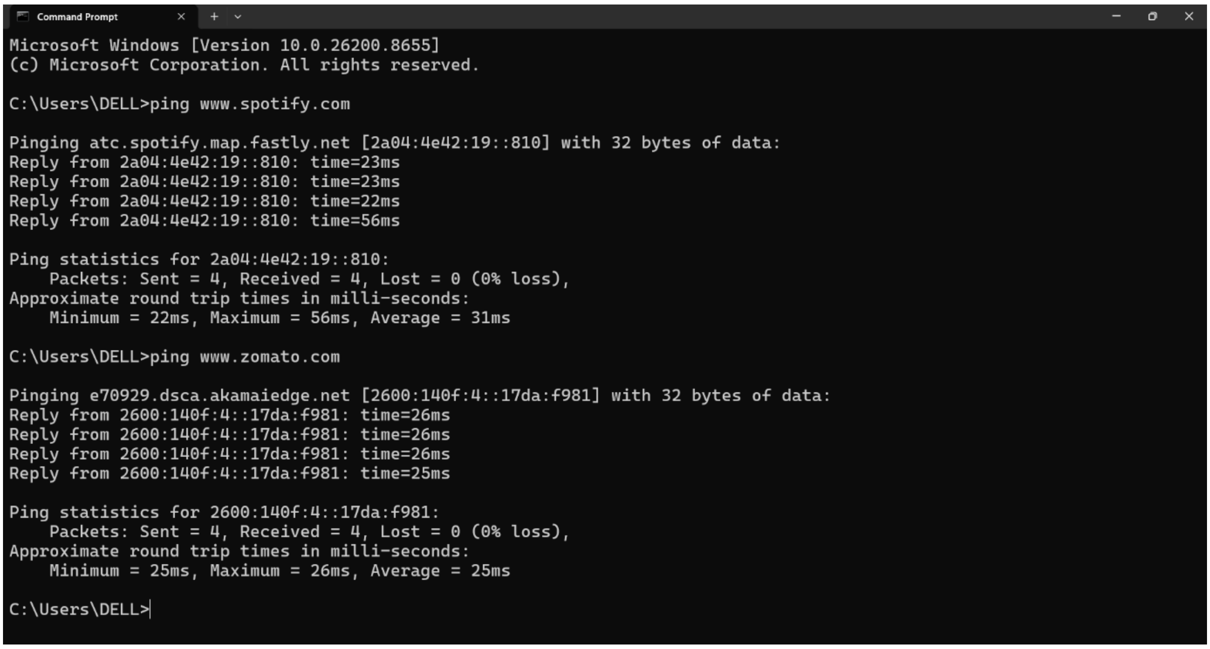
Wireshark captured the network packets generated while browsing YouTube. The **Protocol Hierarchy** feature was used to analyze the captured traffic and identify the most frequently used network protocols during the browsing session.

Conclusion

The network traffic was successfully captured using Wireshark and saved as a **.pcapng** file. The captured data showed that **QUIC**, **TCP**, and **DNS** were the main protocols used while accessing YouTube.

Q2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Ans: Figure 1: Ping Test Results for www.spotify.com and www.zomato.com



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Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:19::810] with 32 bytes of data:
Reply from 2a04:4e42:19::810: time=23ms
Reply from 2a04:4e42:19::810: time=23ms
Reply from 2a04:4e42:19::810: time=22ms
Reply from 2a04:4e42:19::810: time=56ms

Ping statistics for 2a04:4e42:19::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 56ms, Average = 31ms

C:\Users\DELL>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:4::17da:f981] with 32 bytes of data:
Reply from 2600:140f:4::17da:f981: time=26ms
Reply from 2600:140f:4::17da:f981: time=26ms
Reply from 2600:140f:4::17da:f981: time=26ms
Reply from 2600:140f:4::17da:f981: time=25ms

Ping statistics for 2600:140f:4::17da:f981:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 26ms, Average = 25ms

C:\Users\DELL>
```

Response Time Comparison

Website	Minimum Time	Maximum Time	Average Time
www.spotify.com	22ms	56ms	31ms
www.zomato.com	25ms	26ms	25ms

Result

Both Spotify and Zomato were successfully reachable from the computer. **Zomato** responded faster than Spotify because it had a lower average response time (25ms), while Spotify had an average response time of 31ms.

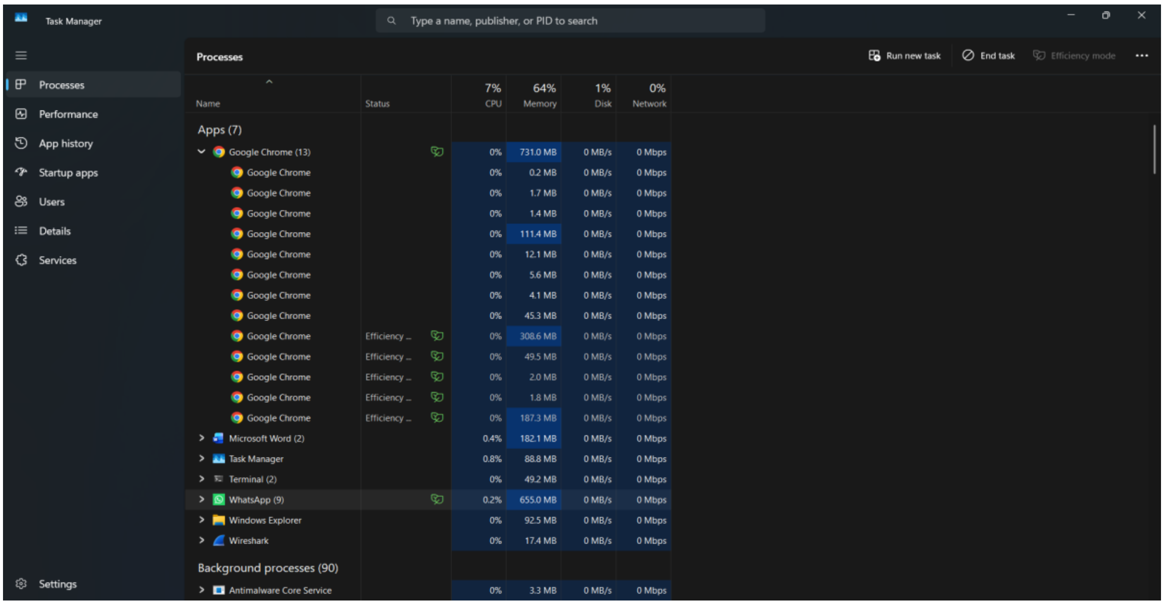
Explanation

The ping command checks whether a website is reachable and measures the time taken for data packets to travel between the computer and the server. A lower average response time indicates faster network communication.

Q3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Ans:

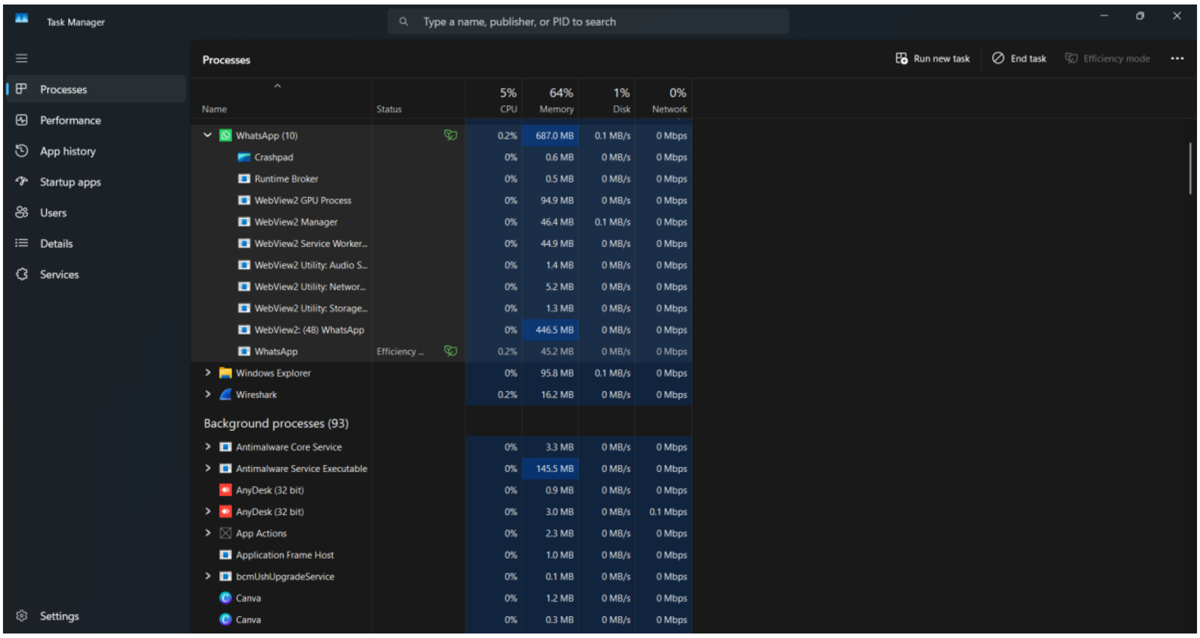
Figure 1: Google Chrome Processes in Task Manager



Google Chrome

Process Name	**Memory Usage**
Google Chrome (Main Process)	**731.0 MB**
Google Chrome Renderer	**308.6 MB**
Google Chrome Renderer	**187.3 MB**
Google Chrome GPU Process	**111.4 MB**
Google Chrome Utility Process	**49.5 MB**

Figure 2: WhatsApp Desktop Processes in Task Manager



WhatsApp Desktop

Process Name	Memory Usage
WhatsApp (Main Process)	**687.0 MB**
WebView2 Process	**448.6 MB**
WebView2 GPU Process	**94.9 MB**
WebView2 Manager	**46.4 MB**
WhatsApp Process	**45.2 MB**

Explanation

Task Manager shows all running applications and their background processes. It also displays the memory used by each process, helping users monitor system performance and identify applications that consume more system resources.

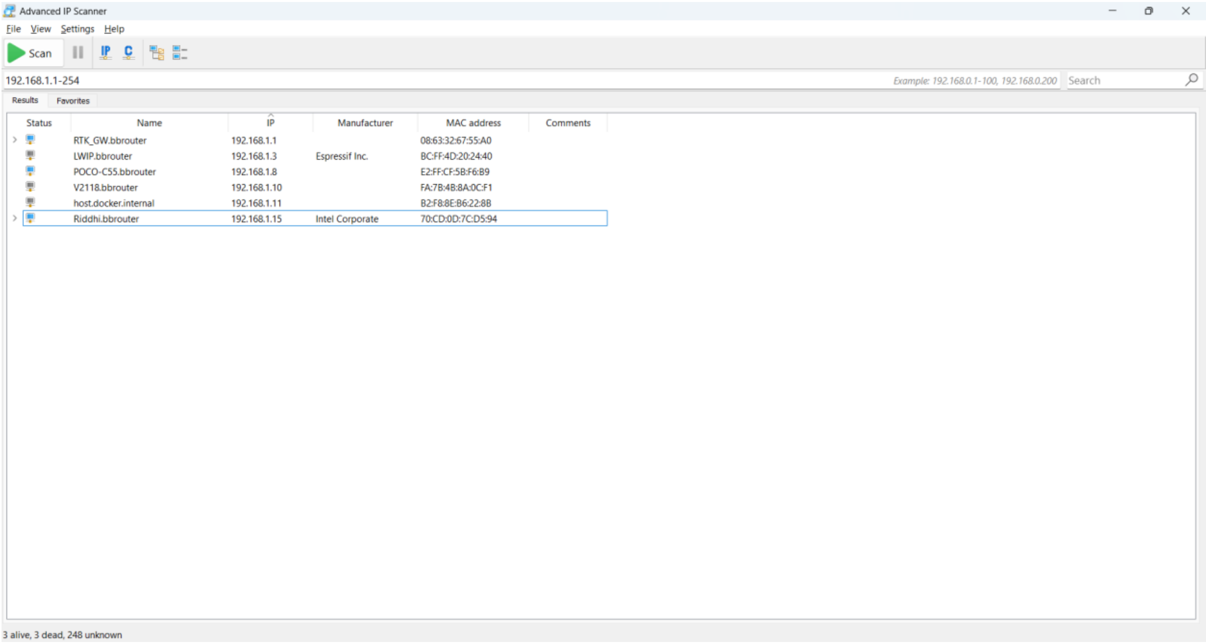
Q4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

Ans:

Procedure

1. Download and install **Advanced IP Scanner**.
2. Open the application.
3. Verify the IP address range.
4. Click the **Scan** button.
5. Wait for the scan to complete.
6. Observe the connected devices.
7. Record the IP address and device name.

Figure 1: Advanced IP Scanner Showing Connected Devices



Connected Devices:

Device Name	IP Address
RTK_GW.bbrouter (Router)	192.168.1.1
POCO-C55.bbrouter	192.168.1.8
Riddhi.bbrouter (Laptop)	192.168.1.15
host.docker.internal	192.168.1.11

Explanation

Advanced IP Scanner scanned the local Wi-Fi network and detected the devices connected to it. The scan displayed useful information such as the IP address, device name, MAC address, and manufacturer.

Q5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Ans: Hardware Tools Used in Network Troubleshooting

Hardware Tool	When It Is Used
Cable Tester	Used to check whether a network (Ethernet) cable is correctly wired or has any faults.
Crimping Tool	Used to attach an RJ45 connector to an Ethernet cable while making or repairing network cables.
Multimeter	Used to measure voltage, current, or continuity when checking power supplies and network equipment.

Explanation

Network technicians use different hardware tools to diagnose and repair network problems. A cable tester checks cable connections, a crimping tool is used to prepare Ethernet cables, and a multimeter helps verify electrical power and continuity in networking devices.

Conclusion

Using the correct hardware tools makes network troubleshooting faster, improves accuracy, and helps maintain a reliable and stable network.